

Editors-in-Chief,
 Journal of Functional Analysis
 Prof. Alexandru D. Ionescu, Prof. Natasa Sesum, and Prof. Stefaan Vaes

Dear Editors-in-Chief of the Journal of Functional Analysis,

I am writing to submit the manuscript “Shorted Schur Complements for Trace-Class Covariance Blocks and Singular Gaussian Conditioning” as an original research article for JFA.

The manuscript is about positive block operators and inverse-free Schur complementation. In finite dimensions, the Schur complement is written as

$$A - BD^{-1}B^*.$$

For compact trace-class Hilbert covariance blocks, this expression is not a valid ambient operator formula: the complementary block has no bounded inverse on the full state space. This paper shows that the correct surviving Schur object is the Anderson–Trapp shorted operator, understood as an inverse-free Schur construction on the Cameron–Martin range.

Our results provide:

1. a rigorous shorting reformulation of the block-triangular OU regularization and rank-one block-operator setup;
2. the affine finite-part scaling law $\Sigma_{\lambda, kk} = \varepsilon_0/(2\lambda^2)$ and residual estimate

$$S_{k,\lambda} = \frac{\varepsilon_0}{2\lambda^2} - R_\lambda, \quad R_\lambda = O(\lambda^{-6}),$$

which separates the singular and finite parts;

3. a Galerkin consistency theory via nested Rayleigh–Ritz truncations, avoiding direct matrix inversion;
4. downstream Gaussian finite-part asymptotics for rank-one disintegration,

$$\mathfrak{J}_\lambda(\mu_{\text{obs}}; \mu_\lambda) = \frac{b_k^2}{\varepsilon_0} + O(\lambda^{-4}),$$

obtained from disintegration and Fredholm determinant/trace cancellations.

Gaussian conditioning enters as the main Radon-law consequence of this operator theorem. Disintegration supplies the exact-evidence measure, while Fredholm determinant/trace cancellations identify the finite part. We also include a concrete 1D elliptic Gaussian field example to demonstrate the framework in a standard trace-class/SPDE setting.

The manuscript is closely aligned with JFA’s scope through its focus on positive block operators, shorted operators, Cameron–Martin geometry, trace-class covariance, and Hilbert-space variational methods; the probabilistic application is a major consequence of the operator-theoretic core.

Thank you for your consideration.

Sincerely,

Lucia Yizi Zhang

Independent Researcher

lucia.zhang.research@outlook.com

<https://orcid.org/0009-0005-3528-7747>